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Correct Answer

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Practice Exam - 1

Question 94 of 117



A 51-year-old woman underwent heart transplantation for non-ischemic dilated cardiomyopathy 4 years ago. She has had no rejection. Her maintenance immunosuppressive drugs include tacrolimus 3 mg twice a day, mycophenolate mofetil (MMF) 1,000 mg twice a day and prednisone 5 mg po daily. She also takes medications for hypertension and hyperlipidemia. She feels fine and stays well hydrated, but her laboratory data shows serum creatinine 2.3mg/dl, which is a new finding. Fasting glucose level is 106 mg/dL with HbA1c 5.6%. Tacrolimus level is 5.1mg/dl. Annual cardiac catheterization shows normal RHC hemodynamics and only mild intimal hyperplasia in the proximal LAD artery by intracoronary ultrasound, not appreciated angiographically.

Which of the following immunosuppressive drug changes would be most appropriate to decrease risk of further renal dysfunction

- ☒ **A.** Add sirolimus and discontinue tacrolimus
- ☐ **B.** Increase prednisone and increase MMF doses
- ☐ **C.** Add sirolimus, discontinue MMF, and decrease tacrolimus dose
- ☐ **D.** Increase tacrolimus dose to suppress cardiac allograft vasculopathy
- ☐ **E.** Continue current regimen and recheck metabolis panel in 3 months

Commentary

References

Testing Point: know side effects of anti-rejection medications and know co-morbidity of heart transplant like renal dysfunction and CAV.

Answer: A. Add sirolimus and discontinue tacrolimus.

This patient has more significant renal dysfunction than cardiac allograft vasculopathy (CAV). Discontinuation of tacrolimus (calcineurin inhibitor) and initiation of sirolimus (mTOR inhibitor) seems to be the most optimal therapeutic move of the choices here.

Worsening renal function occurs commonly in heart transplant patients. Calcineurin inhibitors (CNIs) contribute to renal dysfunction. A transition of CNI to mTOR inhibitor has been shown to slow progression of renal dysfunction. Sirolimus may be a substitute for CNIs to reduce CNI- induced nephrotoxicity.

CAV is a form of chronic allograft rejection and is a major limitation of post-transplant survival. Statin therapy and mTOR inhibitors like sirolimus demonstrated efficacy in slowing progression of CAV. Substitution of calcineurin inhibitors (CNIs) with sirolimus may improve renal function and may also slow down CAV progression. Sirolimus is not indicated for use in early posttransplant course due to increased rejection risk. In this case, the patient is 4 years after heart transplant.

Answers B and D: Increasing prednisone or tacrolimus dose would not help with worsening renal function.

Answer C: Lowering tacrolimus dose is not indicated with the trough level already at the lower limit of therapeutic range.

Answer E: A significant rise in serum creatinine requires prompt intervention and it is not recommended to wait 3 months.

Calcineurin Inhibitors (CNIs): Cyclosporine, Tacrolimus. Proliferation Signal Inhibitors (PSIs) / mechanistic target of rapamycin (mTOR) inhibitors: Sirolimus, Everolimus. Distractors: CAV, different medications

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